	Therefore the vertical component of the velocity calculated **would be greater**.	
6(a)	1.6 J of energy per coulomb means the cell's emf is 1.6 V. This means that the p.d. across the internal resistor is **1.6 – 1.2 = 0.4 V.** Hence the resistance across the internal resistor is **0.4 / 0.5** = 0.8 Ω (shown)	 **[1] working** **[1] working**
(b)	p.d across the bulb Y = $E - Ir$ = 1.6 – (0.3)(0.8) = 1.36 V	 **[1] sub** **[1] ans**
(c)	When connected to bulb Y, Total input power = EI = 1.6 x 0.3 = 0.48 W Output power = VI = 1.36 x 0.3 = 0.408W Power lost = 0.072 W Or Power dissipated in internal resistance = I^2r= $(0.3^2)(0.8)$ = 0.072 W	 **[1] working** **[1] ans**
(d)	Maximum power drawn occurs when $R_Z = r$ (maximum power theorem) Using $E = I\ (R+r)$ 1.6 = I (0.8+0.8) I = 1 A	 **[1] working** **[1] ans**
7(a)	ρL	